INCH-POUND

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SUPERSEDING
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March 25, 1987

COMMERCIAL ITEM DESCRIPTION

TRAMMELS

The General Services Administration has authorized the use of this Commercial Item Description by all federal agencies.

1. SCOPE. This Commercial Item Description covers trammels for layout work and for measuring distance between points too great for ordinary dividers.

2. CLASSIFICATION. The trammels shall conform to the following types.

2.1 Type I, Trammels with steel beam.

2.2 Type II, Trammel heads.

3. SALIENT CHARACTERISTICS.

3.1 Type I Design. Type I trammels shall be of a precision type and consist of a pair of trammel heads, a fine adjustment attachment, a pair of scriber points, a pair of caliper legs, a steel beam, and shall be similar to figure 1.

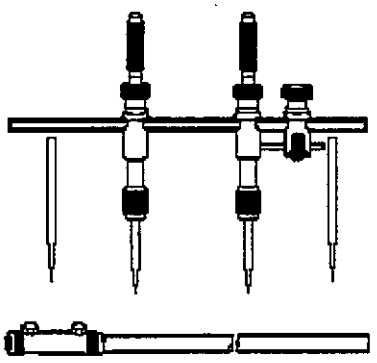


Figure 1. Type I, Trammel with steel beam.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Hardware SuperStore, 6FEE, 1500 Bannister Rd., Bldg.# 6, Kansas City, Missouri. 64131.

FSC 5210

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3.1.1 Trammel heads. One head shall consist of a shaft having a hole to slip fit on the steel beam and a chuck for holding points. A knurled locking nut shall be provided for holding the head securely on the beam. A spring friction device shall be provided to hold the head in position when the nut is loosened. The other head shall be similar, except that a fine adjustment clamp and adjusting screw shall be attached.

3.1.2 Chucks. The chucks shall be of such design as to grip and firmly hold the points at any position within the adjustment range and shall be knurled on the outside diameter.

3.1.3 Locking nuts. Locking nuts shall be round and knurled on the outside diameter.

3.1.4 Fine adjustment clamp. The fine adjustment clamp shall consist of a yoke to accommodate a fine adjustment screw and nut. The clamp shall be made to slip fit on the steel beam and be provided with a locking screw for holding the clamp firmly in position. Clearance between the yoke and bearing surface of nut shall not exceed .005 inch.

3.1.5 Fine adjustment screw and nut. One end of the screw shall pass through the yoke and nut of the fine adjustment clamp. The nut shall be round and knurled on the outside diameter. The other end of the screw shall thread into one of the heads. The screw shall provide a fine adjustment of not less than 3/8 inch.

3.1.6 Scribe points. Trammels shall be furnished with two replaceable scribe points, which shall be made of tool steel. The points shall have a straight shaft with one end tapered from 1 inch to a sharp point. Points shall not be less than 3 inches or more than 5 inches in length. Tip hardness shall be from 50 to 56 on the Rockwell C scale in accordance with ASTM E-18 when tested and have a ground finish.

3.1.7 Caliper legs. Caliper legs shall conform to the requirements for scribe points except the pointed section shall be bent to 90 degrees as the legs of inside calipers.

3.1.8 Beams. Beams shall be made of a steel rod with a flat machined on one side to prevent misalignment when the heads are clamped in position. The beam shall be 5/16 inch, +/- 1/64 inch in diameter, 20 inches, +/- 1/8 inch, in length.

3.2 Type II Design. Type II trammels shall consist of 2 heads with removable scribe points, locking screws, a fine adjustment device, a pencil attachment, and shall be similar to figure 2. (Beam not included).

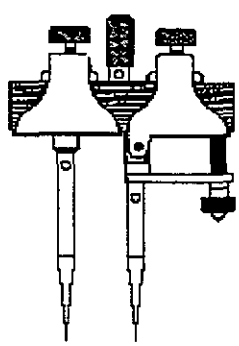


Figure 2. Type II, Trammel heads (beam not included).

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3.2.1 Heads. Trammel heads shall be made of bronze, brass, zinc alloy die casting, or steel. The heads shall be suitable for attaching to a wooden beam 3/4 inch wide and 5/16 inch thick, shall have a locking screw, and suitable for means for attaching points. One head shall have a fine adjustment attachment.

3.2.2 Locking screws. Locking screws shall have a close fit to the threaded hole in the head. The screw head shall be round and knurled. The screw tip shall depress a block or plate to contact the beam as to not leave an indention in the beam and hold the head firmly in place. The block or plate shall be contained in a channel not to turn with the locking screw.

3.2.3 Fine adjustment attachment. A fine adjustment attachment shall be supplied with one of the trammel heads to provide fine adjustment of at least 1/2 inch.

3.2.4 Scribe points. Points shall be made of tool steel with a straight shaft. One end shall be tapered from 1 inch to a sharp point. The opposite end shall be threaded to attach it to the head. Points shall not be less than 2 1/2 inches or more than 5 inches in length. Tip hardness shall be from 50 to 56 on the Rockwell C scale in accordance with ASTM E-18 when tested, and have a ground finish.

3.3 Finish. The trammel shall have a machine ground, chrome, black oxide, or black wrinkle finish.

3.4 Threads. The threads of all components of trammel heads shall be in accordance with Federal Standard H28.

3.5 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be etched, engraved, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.6 Workmanship. Details of workmanship shall be in accordance with the best commercial practices. Paints, coatings, plating, and finishes shall be smooth, adherent, continuous and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g. sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g. pits, rips, fins, burrs, tears, nodules, cracks, or blisters), which may adversely impact the item's serviceability, durability, safety, or appearance

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The supplier/manufacturer is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). .

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch- pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. PRODUCT CONFORMANCE.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the supplier's own drawings, specifications, standards and

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quality assurance practices, and the same product and offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Responsibility for inspection. Unless otherwise specified, the supplier is responsible for the performance of all inspection requirements and may use any commercial facilities (including the supplier's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

6. PACKAGING. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Acquisition Regulation (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20401-9371

GSA Specification Section (3FP-E), Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC, 20407.

American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA, 19428-2959

7.2 National Stock Numbers (NSNs). The following list of NSNs corresponds to the styles specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

Type 1, Trammel with steel beam.

NSN	Type	Details
5210-00-221-1848	I	36-inch long beam.
5210-01-330-1016	I	10 ½-inch long beam.

Type 2, Trammel for rectangular wood-beam (beam not included).

5210-00-221-1850	II	2-heads with points
5210-00-267-4905	II	12-piece set

7.3 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of Commercial Item Description.
(b) Type and details.
(c) When applicable, the appropriate NSN(s)

7.4 Key words. Trammels.

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions and amendments to this Commercial Description until further notice.

Preparing Activity:

GSA - FSS

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